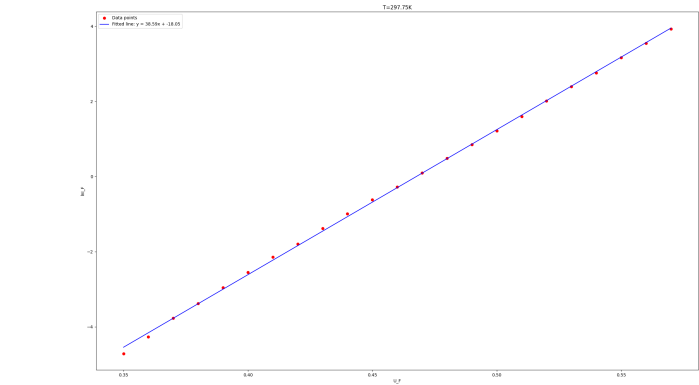


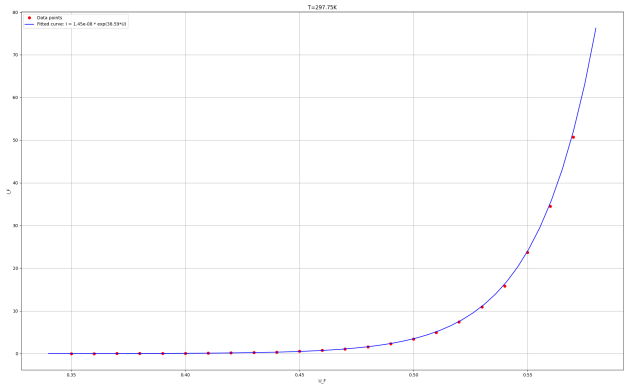
0.1 $U_F - I_F$ 曲线

0.1.1 $T = 297.75K$

序号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
$U_F(V)$	0.350	0.360	0.370	0.380	0.390	0.400	0.410	0.420	0.430	0.440	0.450	0.460	0.470	0.480	0.490	0.500	0.510	0.520	0.530	0.540	0.550	0.560	0.570
$I_F(\mu A)$	0.009	0.014	0.023	0.034	0.052	0.078	0.117	0.166	0.251	0.371	0.538	0.756	1.100	1.620	2.330	3.380	4.940	7.440	10.900	15.800	23.700	34.500	50.700



(a) $\ln I_F = 38.59U_F - 18.05$



(b) $I_F = 1.45 \times 10^{-8} e^{38.59U_F}$

$T = 297.75K$

对于

$I_F = I_s \exp(\frac{qV_F}{kT})$

我们可以得到

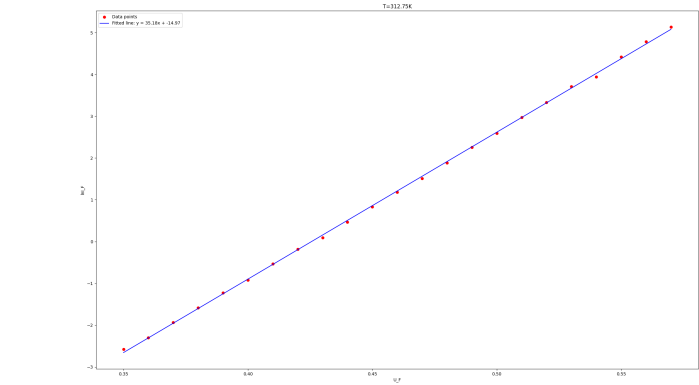
$I_s = 1.45 \times 10^{-8} \mu A, \frac{q}{kT} = 38.59V^{-1}$

我们可以计算出

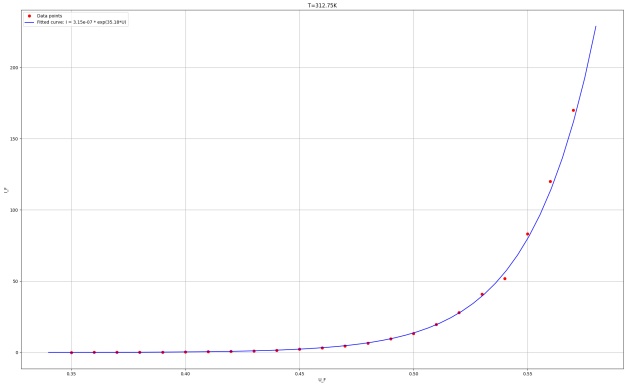
$k = \frac{q}{T \times 38.59V^{-1}} = \frac{1.6 \times 10^{-19}C}{297.75K \times 38.59V^{-1}} = 1.392494325 \times 10^{-23} J/K$

0.1.2 $T = 312.75K$

序号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
$U_F(V)$	0.350	0.360	0.370	0.380	0.390	0.400	0.410	0.420	0.430	0.440	0.450	0.460	0.470	0.480	0.490	0.500	0.510	0.520	0.530	0.540	0.550	0.560	0.570
$I_F(\mu A)$	0.076	0.100	0.144	0.205	0.293	0.399	0.585	0.835	1.100	1.590	2.290	3.270	4.540	6.600	9.560	13.400	19.600	28.000	40.900	51.800	83.300	120.000	170.000



(a) $\ln I_F = 32.9U_F - 12.58$



(b) $I_F = 3.15 \times 10^{-7} e^{35.18U_F}$

$T = 312.75K$

我们可以得到

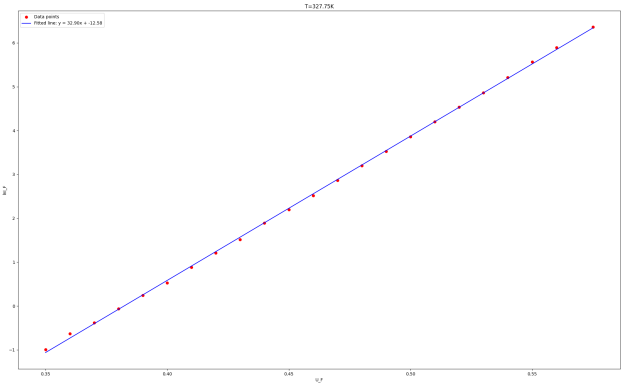
$I_s = 3.15 \times 10^{-7} \mu A, \frac{q}{kT} = 35.18V^{-1}$

我们可以计算出

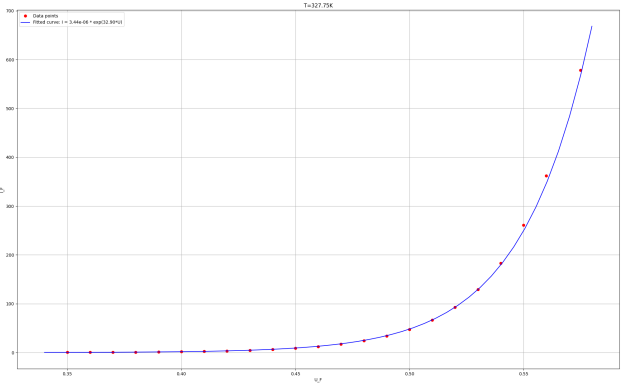
$k = \frac{q}{T \times 35.18V^{-1}} = \frac{1.6 \times 10^{-19}C}{312.75K \times 35.18V^{-1}} = 1.454209003 \times 10^{-23} J/K$

0.1.3 $T = 327.75K$

序号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
$U_F(V)$	0.350	0.360	0.370	0.380	0.390	0.400	0.410	0.420	0.430	0.440	0.450	0.460	0.470	0.480	0.490	0.500	0.510	0.520	0.530	0.540	0.550	0.560	0.564
$I_F(\mu A)$	0.370	0.532	0.680	0.940	1.270	1.690	2.410	3.340	4.560	6.610	8.970	12.400	17.500	24.500	34.000	47.600	66.600	93.100	129.000	183.000	261.000	362.000	578.000



(a) $\ln I_F = 32.90U_F - 12.58$



(b) $I_F = 3.44 \times 10^{-6} e^{32.90U_F}$

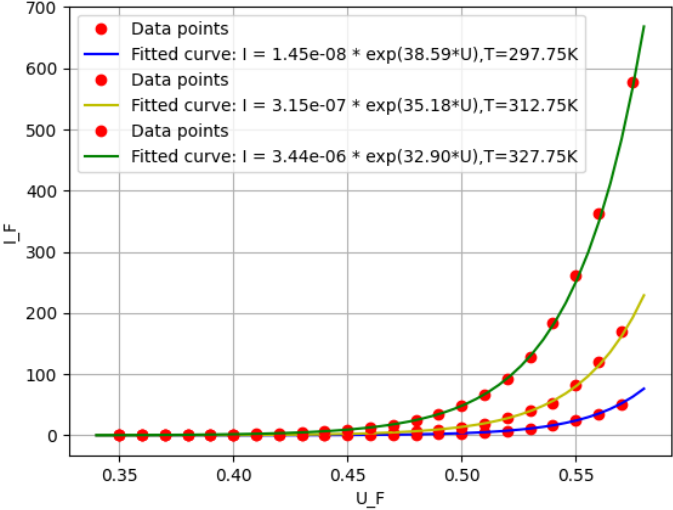
$T = 327.75K$

我们可以得到

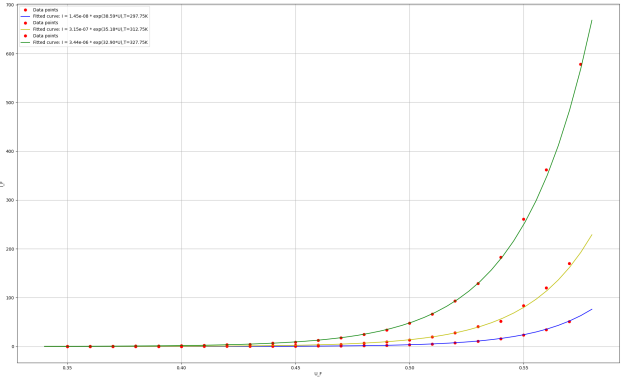
$$I_s = 3.44 \times 10^{-6} \mu A, \frac{q}{kT} = 32.90V^{-1}$$

我们可以计算出

$$k = \frac{q}{T \times 32.90V^{-1}} = \frac{1.6 \times 10^{-19}C}{327.75K \times 32.90V^{-1}} = 1.48382056 \times 10^{-23}J/K$$



(a) 缩略版

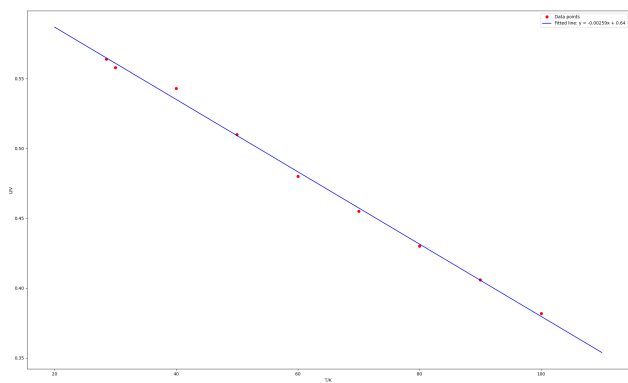


(b) 高清版

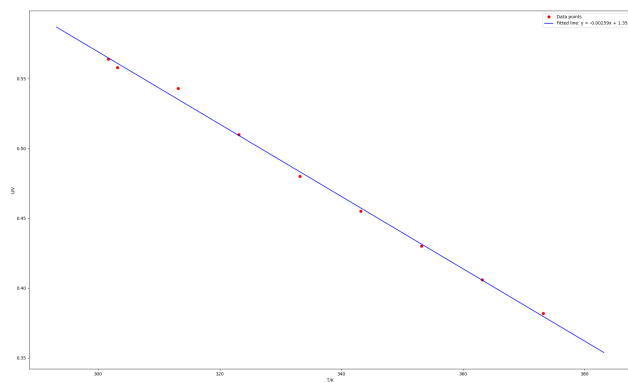
三条曲线

0.2 升温曲线

序号	1	2	3	4	5	6	7	8	9
T/K	301.650	303.150	313.150	323.150	333.150	343.150	353.150	363.150	373.150
$V_F(V)$	0.564	0.558	0.543	0.510	0.480	0.455	0.430	0.406	0.382



(a) 摄氏温标: $V_F = -0.00259T + 0.64$



(b) 热力学温标: $V_F = -0.00259T + 1.35$

升温曲线

我们可以得到

$$S = -\frac{k}{e} \ln \frac{c}{I_F} = -0.00259 \text{ V/K}$$

$$U_g = 1.35 \text{ V}$$

$$E_g = U_g \times 1.6 \times 10^{-19} \text{ C} = 2.16 \times 10^{-19} \text{ V}$$